

Discrete Structure Assignment 01:-

Name:- Yash-Raj

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Q) Question 1:-

Identify the propositional sentences and their truth-values:-

- | | |
|---------------------------|---------------------------|
| (a) Propositional (True) | (b) Propositional (false) |
| (c) Non-propositional | (d) Non-propositional |
| (e) Non-propositional | (f) Non-propositional |
| (g) propositional (false) | (h) propositional (True) |
| (i) propositional (Yes) | (j) Non-propositional |

Q) Question 2:-

Express statements using logical connectives:-

- (a) $P \leftrightarrow Q$
- (b) $P \vee \neg Q$
- (c) $P \vee Q$
- (d) $P \wedge Q$
- (e) $P \oplus Q \quad (P \vee Q) \wedge \neg (P \wedge Q)$

Q Question 3:-

Let p : " $2 \leq 5$ ", q : " 8 is an even integer"

r : " 11 is a prime-number". Express them in english & determine whether true or false.

(a)

$\neg p \wedge q$ (false)

$\neg p$: $2 \geq 5$ (false)

q : 11 is a prime-number (True)

$\neg p \wedge q$: Two is greater than or equals to 5 and 11 is a prime-number
Ans = false

(b) $(p \wedge q) \rightarrow r$

p : $2 \leq 5$ (True)

q : 8 is an even integer (True)

r : 11 is a prime-number (True)

~~$(p \wedge q) \rightarrow r$: Two is less than or equals to five and Eleven is a prime-number, then~~

~~$(p \wedge q) \rightarrow r$: if Two is less than or equals to five and 8 is an even integer then 11 is a prime-number.~~

Ans = True.

(c) $p \rightarrow (q \vee (\neg r))$

$(q \vee (\neg r))$

$\downarrow$
T V F = True.

p : $2 \leq 5$ (True)

q : 8 is an even integer (True)

$\neg r$: 11 is not a prime-number (false)

$p \rightarrow (q \vee (\neg r))$: if two is less than or equals to five then 8 is an even integer or 11 is not a prime-number.

Ans = True.

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(d) $(\neg p) \rightarrow (\neg q)$

$\neg p$: Two is not less than or equals to five (false)

$\neg q$: 8 is not an even integer (false)

$(\neg p) \rightarrow (\neg q)$: if Two is not less than or equals to five, then 8 is not an even integer.

Ans: True

Question 4:-

p : Smartphone A:- 256 RAM, 32 GB, resolution = 8MP

q : Smartphone B:- 288 RAM, 64 GB, resolution = 4MP

r : Smartphone C:- 128 RAM, 32 GB, resolution = 5MP

→ Express statements using logical connectives. Also determine the truth-values of these propositions:-

(a) $(q_{(RAM)} > p_{(RAM)}) \wedge (q_{(RAM)} > r_{(RAM)})$ Ans:- True

(b) $(r_{(RAM)} > q_{(RAM)}) \vee (r_{(resolution)} > q_{(resolution)})$ Ans:- True

(c) $(q_{(RAM)} > p_{(RAM)}) \wedge (q_{(RAM)} > p_{(RAM)}) \wedge (q_{(resolution)} > p_{(resolution)})$ Ans:- False

(d) $(q_{(RAM) \times (RAM)} > r_{(RAM) \times (RAM)}) \rightarrow (q_{(resolution)} > r_{(resolution)})$ Ans:- False

(e) $(p_{(RAM)} > q_{(RAM)}) \leftrightarrow (q_{(RAM)} > p_{(RAM)})$ Ans:- False

Question 5:-

P: Acme-computers Revenue: 138B, Net-profit: 8 billion

Q: Nadir-software Revenue 87B, Net-profit: 5 billion

R: Quixote Media Revenue 111B, Net-profit: 13 billion

→ Express them using logical-connectives, and also their truth-values:-

$$(a) (Q_{(Revenue)} > N_{(Revenue)}) \wedge (Q_{(Revenue)} > A_{(Revenue)})$$

$$(b) (Q_{(profit)} < P_{(profit)}) \wedge (Q_{(profit)} < R_{(profit)}) \wedge (P_{(Revenue)} > Q_{(Revenue)}) \wedge (P_{(Revenue)} > R_{(Revenue)})$$

$$(c) [(P_{(profit)} > Q_{(profit)}) \wedge (P_{(profit)} > R_{(profit)})] \vee [(R_{(profit)} > Q_{(profit)}) \wedge (R_{(profit)} > P_{(profit)})]$$

$$(d) [(R_{(profit)} < Q_{(profit)}) \wedge (R_{(profit)} < P_{(profit)})] \rightarrow [(P_{(Revenue)} > Q_{(Revenue)}) \wedge (P_{(Revenue)} > R_{(Revenue)})]$$

$$(e) [(Q_{(profit)} < P_{(profit)}) \wedge (Q_{(profit)} < R_{(profit)})] \leftrightarrow [(P_{(Revenue)} > Q_{(Revenue)}) \wedge (P_{(Revenue)} > R_{(Revenue)})]$$

$$(a) T \wedge F \text{ is false}$$

$$(b) T \wedge T \wedge T \wedge T \text{ is True}$$

$$(c) [T \wedge F] \vee [T \wedge T] \text{ is True}$$

$$(d) [F \wedge F] \rightarrow [T \wedge T] \text{ is True}$$

$$(e) [T \wedge F] \leftrightarrow [T \wedge T] \text{ is True}$$

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Q Question 6:-

P : You have the flu.

Q : You miss the final examination.

r : You pass the course.

→ Express each of the following as English sentence.

(a) $P \rightarrow Q$:- If you have the flu then miss the final examination.

(b) $\neg Q \leftrightarrow r$:- You don't miss the final examination if and only if you pass the course.

(c) $Q \rightarrow \neg r$:- If you miss the final examination, then you don't pass the course.

(d) $P \vee Q \vee r$:- You have flu or You miss the final examination or you pass the course.

(e) $(P \rightarrow \neg r) \vee (Q \rightarrow \neg r)$:- if you have the flu, then you don't pass the course or if you miss the final examination, then you don't pass the course.

(f) $(P \wedge Q) \vee (\neg Q \wedge r)$:- You have the flu and you miss the final examination or You don't miss the examination and you pass the course.

Q) Question 7:-

⇒ Conjunction (and, but, ...)
nevertheless

p : You get an A on the final Exam.

q : You do every exercise in this book.

r : You get an A in the class.

→ Write these propositions using p, q and r and logical-connectives:-

(a) $r \wedge \neg q$

(b) $p \wedge (q \wedge r)$

(c) $r \rightarrow p$

(d) $(p \wedge \neg q) \wedge r$

(e) $(p \wedge q) \rightarrow r$

(f) $r \leftrightarrow (q \vee p)$

Q) Question 8:- Write statements in the p, q form:

(a) If you send me an e-mail message then I will remember to send you the address.

(b) If you were born in the United States, then you are a citizen of this country.

(c) If you keep your textbook, then it will be a useful reference in your future.

(d) If their goalie plays well, then the red wings will win Stanley cup.

(e) If you get the job, then you have the best credentials.

(f) If there is a storm, then beach closes.

(g) If you can log on to the server, then you have a valid password.

(h) If you don't begin your climb too late, then you will reach the summit.

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⇒ Question:- 9:

→ if it is ~~necessary~~ sunny tomorrow, then i will go for a walk in the woods.

(a) at least 5 ways to write $P \rightarrow Q$ in English:-

(i) I will go for a walk in the woods whenever it is sunny tomorrow.

(ii) I will go for a walk in the woods if it is sunny tomorrow

(iii) for me to go for a walk in the woods tomorrow, it is sufficient that it is sunny

(iv) Unless it is not sunny tomorrow, I will go for a walk in the woods

(v) It is sunny tomorrow only if i will go for a walk in the woods

(b) State converse, inverse and contrapositive of this statement:-

(i) Converse ($Q \rightarrow P$): If i go for a walk in the woods, then it is sunny tomorrow.

(ii) Inverse ($\neg P \rightarrow \neg Q$): If it is not sunny tomorrow then it will not go for a walk in the woods.

(iii) Contrapositive ($\neg Q \rightarrow \neg P$): If i don't go for a walk then it is not sunny tomorrow.

(c) find inverse of inverse, converse & contrapositive:

(i) Inverse of inverse $[\neg(\neg p \rightarrow \neg q)] = (p \rightarrow q)$: if it is sunny tomorrow, then i will go for a walk in the woods.

(ii) Inverse of converse $[\neg(q \rightarrow p)] = (\neg q \rightarrow \neg p)$: if it is not sunny tomorrow then i will not go for a walk in the woods.

(iii) Inverse of contrapositive $[\neg(\neg q \rightarrow \neg p)] = (q \rightarrow p)$: If i go for a walk in the woods then it is sunny tomorrow.

Q) Question 14:-

$$q(x, y) = x + y = 1$$

→ which of the following are propositions & which are not & also determine their truth-values.

(a) $q(x, y)$
 $q(x, y) = x + y = 1$
→ Non-proposition

(b) $q(-6, 7)$
 $q(-6, 7) = -6 + 7 = 1$
proposition (True)

(c) $q(x+1, -x)$
 $q(x+1, -x) = x+1-x = 1$
1 = 1
Not a proposition

(d) $q(x, 3)$
 $q(x, 3) = x + 3 = 1$
Not a proposition

(e) $q(1, 1)$
 $q(1, 1) = 1 + 1 = 1$
1 + 1 = 1
→ Non-proposition (false)

(f) $q(5, -4)$
 $q(5, -4) = 5 - 4 = 1$
→ proposition (True)

Discrete Assignment #1

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Question #10:-

Use de-morgan law to find the Negation of the following:-

(a) Jan is rich & happy.

Soln.

$\therefore P \wedge Q$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

Ans. Jan is not rich or not happy.

(b) Carlos will bicycle or run tomorrow.

Soln.

$\therefore (P \vee Q)$

$$\therefore \neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

Ans. Carlos will not bicycle and he will not run tomorrow.

(c) The fan is slow or it is very hot.

Soln.

$\therefore (P \vee Q)$

$$\therefore \neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

Ans. The fan is not slow and it is not very hot.

(d) Akram is unfit & Ahmed is injured.

Soln.

$\therefore (P \wedge Q)$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

Ans. Akram is ~~not~~ fit & Ahmed is not injured.

(e) $0 \leq x \leq 4$ ($x \geq 0$) & ($x \leq 4$)
 $\underline{\text{Soln.}}$
 $(P \wedge Q)$
 $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

(f) $-2 < x \leq 5$ ($x > -2$)
 $\underline{\text{Soln.}}$ ($x \leq 5$)
 $(P \wedge Q) \rightarrow \neg(P \wedge Q) \equiv \neg P \vee \neg Q$
Ans $(x \leq -2) \vee (x > 5)$

Ans $4 \leq x \leq 0$
 $(x < 0) \vee (x > 4)$ Ans.

(g) The fan is slow, or it is very hot.
 $\underline{\text{Soln.}}$
 $(P \vee Q) \Rightarrow \neg(P \vee Q) \equiv \neg P \wedge \neg Q$

Ans The fan is not and it is not very hot.

(h) Akram is fit & Salim is injured.
 $\underline{\text{Soln.}}$
 $(P \wedge Q) \Rightarrow \neg(P \wedge Q) \equiv \neg P \vee \neg Q$

Ans Akram is not fit & Salim is not injured.

Question # 11:-

- (a) Exclusive OR
- (b) Inclusive OR
- (c) Inclusive OR
- (d) Inclusive OR.

Question #13:-

Using truth-table, show that these compound propositions are logically equivalent or not:-

(a) $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$:-

Soln :-

p	q	r	$p \rightarrow r$	$q \rightarrow r$	$p \vee q$	$(p \rightarrow r) \wedge (q \rightarrow r)$	$(p \vee q) \rightarrow r$
0	0	0	1	1	0	1	1
0	0	1	1	1	0	1	1
0	1	0	1	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	0	0
1	0	1	1	1	1	1	1
1	1	0	0	0	1	0	0
1	1	1	1	1	1	1	1

Ans:- $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ both are logically equivalent.

(b) $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$:-

p	q	r	$(p \rightarrow q)$	$p \rightarrow r$	$q \vee r$	$(p \rightarrow q) \vee (p \rightarrow r)$	$p \rightarrow (q \vee r)$
0	0	0	1	1	0	1	1
0	0	1	1	1	1	1	1
0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0
1	0	1	0	1	1	1	1
1	1	0	1	0	1	1	1
1	1	1	1	1	1	1	1

Ans:- These both are logically equivalent.

(c) $(P \rightarrow Q) \rightarrow (R \rightarrow S)$ and $(P \rightarrow R) \rightarrow (Q \rightarrow S)$:

P	Q	S	R	$P \rightarrow Q$	$R \rightarrow S$	$P \rightarrow R$	$Q \rightarrow S$	$(P \rightarrow Q) \rightarrow (R \rightarrow S)$	$(P \rightarrow R) \rightarrow (Q \rightarrow S)$
0	0	0	0	1	1	1	1	1	1
0	0	0	1	1	0	1	1	0	1
0	0	1	0	1	1	1	1	1	1
0	0	1	1	1	1	1	1	1	1
0	1	0	0	1	1	1	0	1	0
0	1	0	1	1	0	1	0	0	0
0	1	1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1
1	0	0	0	0	1	0	1	1	1
1	0	0	1	0	0	1	1	1	1
1	0	1	0	0	1	0	1	1	1
1	0	1	1	0	1	1	1	1	1
1	1	0	0	1	1	0	0	1	1
1	1	0	1	1	0	1	0	0	0
1	1	1	0	1	1	0	1	1	1
1	1	1	1	1	1	1	1	1	1

Ans: These both are not equivalent.

Question #15:-

- (a) $P(4,5)$ is false
- (b) $P(2,4)$ is True.
- (c) $\forall m \forall n P(m,n)$ is false.
- (d) $\exists m \forall n P(m,n)$ is True.
- (e) $\exists n \forall m P(m,n)$ is false.
- (f) $\forall n P(1,n)$ is True.

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Question 16:- Determine the truth-values

(a) $\exists x(x^2=2)$:- This says that there exist some values of x for which $x^2=2$. This statement is True. Since, $x^2=2 \Rightarrow x=\pm\sqrt{2}$ which is a real value.

(b) $\exists x(x^2=-1)$:- This says that there exist some values of x for which $x^2=-1$. This statement is false, since $x=\pm\sqrt{-1}$ is not a real value.

(c) $\forall x(x^2+2 \geq 1)$:- This is a true statement as x can be anything and answer will be the real-number.

(d) $\exists x(x^2=x)$:- This statement is true. Since, $x^2=x$
 $\Rightarrow x=\pm\sqrt{x}$
Both are real values

Question #17:

$F(x, y)$ x can fool y where domain consists of all people in the world.

(a) $\forall x F(x, Bob)$

(b) $\forall y F(Alice, y)$

(c) $\forall x \exists y F(x, y)$

(d) $\neg \exists x \forall y F(x, y)$

(e) $\forall y \exists x F(x, y)$

~~(f)~~

Question #18: $P(x)$ x can speak Russian

$Q(x)$ x knows the computer language C++.

domain consists of all students at your school.

(a) $\exists x (P(x) \wedge Q(x))$

(b) $\exists x (P(x) \wedge \neg Q(x))$

(c) $\forall x (P(x) \vee Q(x))$

(d) $\forall x (\neg P(x) \wedge \neg Q(x))$

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Question #19:-

$Q(x,y)$ where domain is all student in your class.

statement:- x has sent an e-mail message to y .

- (a) $\exists x \exists y Q(x,y)$:- This statement says that:- There exist two student x and y such that x has sent e-mail to y .
- (b) $\exists x \forall y Q(x,y)$:- This says that There exist some student (or a student) who has sent an email to everyone.
- (c) $\forall x \exists y Q(x,y)$:- This says that:- Everyone has sent an email to atleast one student.
- (d) $\exists y \forall x Q(x,y)$:- This says:- There is a student who has received from every student in the class.
- (e) $\forall y \exists x Q(x,y)$:-
This says:- Someone has sent an email to Everyone.

(A) $\forall x \forall y Q(x,y)$: Everyone has e-mailed everyone.

Question #20:

$P(x,y)$ by the statement "student x has taken class y " where domain for x is "all students in your class" and for y is "all cs courses at your school".

(a) $\exists x \exists y P(x,y)$: This says that: There exists some x for some y and in such a way that: There is at least one student in the class who has taken at least one cs course at your school.

(b) $\exists x \forall y P(x,y)$: There is at least one student in the class who has taken all cs courses at your school.

(c) $\forall x \exists y P(x,y)$: Everyone in the class has taken at least one cs course at your school.

(d) $\exists y \forall x P(x,y)$: At least one cs course at your school has been taken by all the students in your class.

(e) $\forall y \exists x P(x,y)$: Every cs course has been taken by at least one student.

(f) $\forall x \forall y P(x,y)$: Every student in the class has taken all cs courses at your school.

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Question #21::- What rule of inference is used in this?

(a) $P \therefore P \vee Q$ (Addition)

(b) $P \wedge Q \therefore P$ (simplification)

(c) $P \rightarrow Q \therefore Q$ (modus Ponens)

(d) $P \rightarrow Q$ (modus Tollens)

(e) $(P \rightarrow Q) (Q \rightarrow R) \therefore P \rightarrow R$ (Hypothetical syllogism)

Question #22:-

(a) $T \rightarrow (M \vee E)$ (premise)

$S \rightarrow \neg E$ (premise)

$T \wedge S$ (premise)

T

S

$M \vee E$

$\therefore$

$\neg E \rightarrow M$ (conclusion)

(b) $P \rightarrow Q$ (premise) (1) $\neg Q \rightarrow \neg P$

$r \rightarrow \neg s$ (premise) (2) r from $Q \rightarrow r$ (premise)

$Q \rightarrow r$ (premise) (3) $\neg s$ (4) $P \rightarrow \neg s$ $\therefore$

(c) $(E \rightarrow (r \vee p)) \rightarrow ((\neg r \vee k) \wedge \neg k) \rightarrow \neg r$

(1) r

(2) Assume E

(3) $r \vee p$

(4) $E \rightarrow (r \vee p)$

(5) $(\neg r \vee k) \wedge \neg k$

(6) $\neg r \vee k$

(7) $\neg k$

(8) k

(9) $k \wedge \neg k$

(10) $\neg r$

(11) $((E \rightarrow (r \vee p)) \rightarrow ((\neg r \vee k) \wedge \neg k) \rightarrow \neg r$

Q#29:-

$$f(a) = 2a + 3$$

$$g(a) = 3a + 2$$

(a) $f \circ g(a) = 6a + 7$, $g \circ f(a) = 6a + 11$

(b) Both are linear, injective, not surjective.

(c) They are not invertible. *Pro.*

Q#30:- Real world applications:-

(a) propositional logic:-

- (i) Digital Circuit Design:- propositional logic is the backbone of the digital electronics. for ex:- microprocessors use propositional logic to perform decision making.
- (ii) Software Testing and verification:- It is used to check whether a program satisfies certain conditions. for ex:- user input wrong it denies come under this.

(b) Predicates & Quantifiers:-

- (i) Databases (SQL):- Quantifiers are used to express conditions like:- for all customers ($\forall$) or There exist a customer ($\exists$) etc.
- (ii) Artificial Intelligence (AI):- predicates & quantifiers are used in knowledge representation.

(c) Sets:-

- (i) Data classification:- Sets are used in grouping the data making the programmer (python, R etc) work easy.
- (ii) Probability and statistics:- In probability, set represent events & calculate probability depended on Union, intersection and complement of set.

(d) functions:-

- (i) Computer science:- functions are used to map the input & outputs for example:- Hashing and mapping.
- (ii) Real model modeling:- functions model relationships.

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Question 28:-

(a) $f(n) = \lfloor \frac{n^2}{3} \rfloor$, find $f(s)$ if:-

(i) $S = \{-2, -1, 0, 1, 2, 3\}$

$$\lfloor \frac{(-2)^2}{3} \rfloor = \lfloor \frac{4}{3} \rfloor = 1$$

$$\lfloor \frac{(-1)^2}{3} \rfloor = \lfloor \frac{1}{3} \rfloor = 0$$

$$\lfloor \frac{0}{3} \rfloor = 0, \quad \lfloor \frac{(2)^2}{3} \rfloor = 1$$

$$\lfloor \frac{1}{3} \rfloor = 0, \quad \lfloor \frac{(3)^2}{3} \rfloor = 3$$

$$f(s) = \{0, 1, 3\}$$

(ii) $S = \{0, 1, 2, 3, 4, 5\}$

$$\lfloor \frac{(0)^2}{3} \rfloor = 0$$

$$\lfloor \frac{1}{3} \rfloor = 0$$

$$\lfloor \frac{4}{3} \rfloor = 1$$

$$\lfloor \frac{9}{3} \rfloor = 2$$

$$\lfloor \frac{16}{3} \rfloor = 3$$

$$\lfloor \frac{25}{3} \rfloor = 8$$

$$f(s) = \{0, 1, 2, 3, 8\}$$

(iii) $S = \{1, 5, 7, 11\}$

$$\lfloor \frac{1}{3} \rfloor = 0$$

$$\lfloor \frac{25}{3} \rfloor = 8$$

$$\lfloor \frac{49}{3} \rfloor = 16$$

$$\lfloor \frac{121}{3} \rfloor = 40$$

$$f(s) = \{0, 8, 16, 40\}$$

(iv) $S = \{2, 6, 10, 14\}$

$$\lfloor \frac{4}{3} \rfloor = 1$$

$$\lfloor \frac{36}{3} \rfloor = 12$$

$$\lfloor \frac{100}{3} \rfloor = 33$$

$$\lfloor \frac{196}{3} \rfloor = 65$$

$$f(s) = \{1, 12, 33, 65\}$$

(b) (i) $\lfloor \frac{3}{4} \rfloor$

$$\lfloor \frac{3}{4} \rfloor = 0.75 \rightarrow 1 \text{ Ans}$$

(ii) $\lfloor \frac{7}{8} \rfloor$

$$\lfloor \frac{7}{8} \rfloor = 0$$

(iii) $\lfloor -\frac{3}{4} \rfloor$

$$\lfloor -\frac{3}{4} \rfloor = 0$$

(iv) $\lfloor -\frac{7}{8} \rfloor$

$$\lfloor -\frac{7}{8} \rfloor = -0.875$$

$$= -1 \text{ Ans}$$

(v) $\lfloor 3 \rfloor$

$$3 \text{ Ans}$$

(vi) $\lfloor -1 \rfloor$

$$-1 \text{ Ans}$$

(vii) $\lfloor \frac{1}{2} + \lfloor \frac{3}{2} \rfloor \rfloor$

$$\lfloor 0.5 + 2 \rfloor = 2.5$$

$$2 \text{ Ans}$$

(viii) $\lfloor \frac{1}{2} \cdot \lfloor \frac{5}{2} \rfloor \rfloor$

$$\lfloor 0.5 \cdot 2 \rfloor$$

$$\lfloor 1 \rfloor = 1 \text{ Ans}$$

(c) prove/disprove :-

$$(i) \lfloor -x \rfloor = -\lceil x \rceil$$

Soln

$\lceil x \rceil$ is the smallest integer $\geq x$
so if $\lceil x \rceil = n$ then

$$n-1 < x \leq n$$

multiply -1 :-

$$-(n-1) > -x \geq -n$$

$$\Rightarrow -x \leq -n < -(n-1)$$

$$\lfloor -x \rfloor \leq -n < -\lceil x \rceil$$

Proved $\square$

$$(ii) \lceil -x \rceil = -\lfloor x \rfloor$$

Soln $\lfloor x \rfloor = m$ means

$$m \leq x < m+1$$

multiply -1

$$-m \geq -x > -(m+1)$$

$$\Rightarrow -(m+1) < -x \leq -m$$

$$\lceil -x \rceil \leq -m = -\lfloor x \rfloor$$

Proved $\square$

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Question #24:-

(a) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$

 $A \subseteq B$ Yes every element of A is in B $A \subset B$ Yes $A \neq B$ as $4 \in B$ & not $\in A$ so, $A \subset B$

(b) $\{x/n \text{ is a prime } < 10\} = \{2, 3, 5, 7\}$

$\mathcal{P} = \{\emptyset, \{2\}, \{3\}, \{5\}, \{7\}, \{2, 3\}, \{2, 5\}, \{2, 7\}, \{3, 5\}, \{3, 7\}, \{5, 7\},$

$\{2, 3, 5\}, \{2, 3, 7\}, \{2, 5, 7\}, \{3, 5, 7\}\}$

(c) $L = \{1, 2, 3\}$

$2^3 = 8$ (All-subsets)

$\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\} \rightarrow \text{power-set.}$

(d)

if M has k elements, then $|P(M)| = 2^k$

Each element may be either included or excluded from a subset.

(e) $\{11, 13, 17, 19\}$

power-set = $2^4 = 16$ Ans.

(f)

$E = \{1, 2, 4\}$, $F = \{a, b, c, d\}$

find $E \times F$?

$E \times F = \{(1, a), (1, b), (1, c), (1, d), (2, a), (2, b), (2, c), (2, d), (4, a), (4, b), (4, c), (4, d)\}$

Ans.

Question #26

(a) Let A apples with amount 100/20

Let B apples with amount 100/10

Let C apples with amount 100/10

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$20 + 10 = 10 + 20$$

$$100 - 20 = 80 \rightarrow \text{apples to be sold}$$

(b)

Total = 1000

$$CS = 350, SE = 450, |CS \cap SE| = 100$$

$$|CS \cup SE| = 350 + 450 - 100$$

$$= 700$$

$$1000 - 700 = 300 \text{ Mrs.}$$

(c)

mixed berry = 75

$$|MB \cap ILAT| = 5$$

Just cream = 32

$$None = 14$$

Tramcar = 17

(b) MC and IL = 13

$$|MB \cup ILAT| = (75 + 32 + 17) - (13 + 21 + 16) + 14$$

$$= 122 + 14 = 136$$

Total Mrs

(c) T and MB = 16

$$(d) \text{Set-builder} = \{(x, y) | x \in A, y \in B \cap C\} = \{(x, y) | x \in A, y \in B, y \in C\} = \{(x, y) | x \in A, y \in B\} \cap \{(x, y) | x \in A, y \in C\}$$

→ Element-wise:

Take an ordered pair (x, y)

→ $x \in A$ and $y \in (B \cap C)$

→ $x \in A$ and ($y \in B$ and $y \in C$)

→ ($x \in A$ and $y \in B$) and ($x \in A$ and $y \in C$)

→ $(x, y) \in A \times B$ and $(x, y) \in A \times C$

→ $(x, y) \in (A \times B) \cap (A \times C)$

(x, y) belongs to the $(A \times B) \cap (A \times C)$ Hence

Both sets are equal.

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Question #27:

$$A = \{a, b, c, d\}, B = \{a, b, c, d\}$$

$$(a) f(a) = b, f(b) = a, f(c) = c, f(d) = d$$

- (i) Range: $\{a, b, c, d\}$ {Every element of B appears as an image?}
- (ii) Injective: Yes (All outputs are distinct)
 Surjective: Yes (Range = co-domain)
 Bijective: Yes

$$(iii) \text{ Inverse Exists (Self-Inverse) } \Rightarrow f^{-1}(a) = b, f^{-1}(b) = a, f^{-1}(c) = c, f^{-1}(d) = d$$

$$(b) f(a) = b, f(b) = b, f(c) = d, f(d) = c$$

- (i) Range = $\{b, c, d\}$
- (ii) Injective: No (No distinct outputs).
 Surjective: No (a is not in the range)
 Bijective: No
- (iii) Inverse not possible.

$$(c) f(a) = d, f(b) = b, f(c) = c, f(d) = d$$

$$(i) \text{ Range } = \{b, c, d\}$$

- (ii) Injective: No
 Bijective: No
 Surjective: No

$$(iii) \text{ No inverse}$$

(d) $f(a) = c, f(b) = a, f(c) = b, f(d) = d$
Soln.

(i) Range = $\{a, b, c, d\}$

(ii) Injective: Yes

Bijective: Yes

Surjective: Yes

(iii) Inverse exists & it is:

$f^{-1}(a) = b, f^{-1}(b) = c, f^{-1}(c) = a, f^{-1}(d) = d$

Question #12:-

(a) $(p \wedge (\neg(\neg p \vee q))) \vee (p \wedge q) \equiv p$

Soln.
 $p \wedge (\neg(\neg p \vee q)) \vee (p \wedge q) \equiv p$

$\underbrace{(p \wedge (\neg(\neg p \vee q)))}_p \vee \underbrace{(p \wedge q)}_q$

$(p \wedge p \wedge \neg q) \vee (p \wedge p \wedge q)$

$(p \wedge \neg q) \vee (p \wedge q)$

$p \wedge (\neg q \vee q)$

$p \wedge t$

$p \equiv p$ Proved.

(b) $\neg(p \leftrightarrow q) \equiv (p \leftrightarrow \neg q)$

Soln.
 $\neg(p \leftrightarrow q) \equiv \neg[(p \wedge q) \vee (\neg p \wedge \neg q)]$

$\neg(p \wedge q) \wedge \neg(\neg p \wedge \neg q)$

$(\neg p \vee \neg q) \wedge (p \vee q)$

$p \leftrightarrow \neg q \equiv p \leftrightarrow \neg q$ Proved.

Question 12 ::

(c) $\neg p \leftrightarrow \xi \equiv p \leftrightarrow \neg \xi$

Soln.

$$\begin{aligned} p \leftrightarrow \xi &= (p \wedge \xi) \vee (\neg p \wedge \neg \xi) \\ \neg p \leftrightarrow \xi &= (\neg p \wedge \xi) \vee (p \wedge \neg \xi) \\ p \leftrightarrow \neg \xi &= (p \wedge \neg \xi) \vee (\neg p \wedge \xi) \\ &\text{proved} \end{aligned}$$

(d) $(p \wedge \xi) \rightarrow (p \rightarrow \xi) \equiv T$

Soln.

$$\begin{aligned} (p \wedge \xi) \rightarrow (p \rightarrow \xi) &\equiv T \\ \neg(p \wedge \xi) \vee (\neg p \vee \xi) \\ &= \neg p \vee \neg \xi \vee \neg p \vee \xi \\ &= \neg p \vee T \\ &= T \equiv T \text{ proved.} \end{aligned}$$

(e) $\neg(p \vee \neg(p \wedge \xi)) \equiv F$

Soln.

$$\begin{aligned} \neg(p \vee \neg(p \wedge \xi)) &\equiv F \\ \neg(T \vee \neg \xi) \\ \neg(T) &\equiv F \\ C &\equiv F \\ &\text{proved} \end{aligned}$$

Q.E.D.

Question #23 :-

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A = \{1, 2, 4, 5\}$$

$$B = \{2, 3, 5, 6\}$$

$$C = \{4, 5, 6, 7\}$$

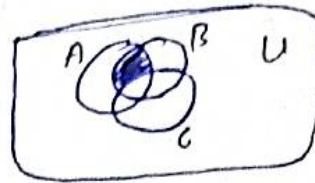
(i) $(A \cap B) \cap \bar{C}$

soln

$$A \cap B = \{2, 5\}$$

$$\bar{C} = U - C = \{1, 2, 3, 8\}$$

$$(A \cap B) \cap \bar{C} = \{2\}$$



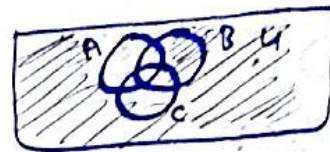
(ii) $\bar{A} \cup (B \cap C)$

soln

$$B \cap C = \{2, 3, 4, 5, 6, 7\}$$

$$\bar{A} = U - A = \{3, 6, 7, 8\}$$

$$\bar{A} \cup (B \cap C) = \{2, 3, 4, 5, 6, 7, 8\}$$

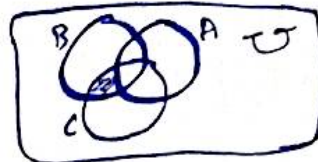


(iii) $(A - B) \cap C$

soln

$$A - B = \{1, 4\}$$

$$(A - B) \cap C = \{4\}$$



(iv) $(A \cap \bar{B}) \cup \bar{C}$

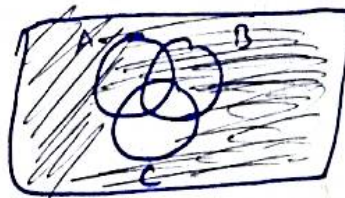
soln

$$B' = \{1, 4, 7, 8\}$$

$$C' = \{1, 2, 3, 8\}$$

$$A \cap \bar{B} = \{4\}$$

$$(A \cap \bar{B}) \cup \bar{C} = \{1, 2, 3, 4, 8\}$$



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Question #25 ::

$$(a) (A - (A \cap B)) \cap (B - (A \cap B)) = \emptyset$$

Soln.

$$(A \cap (\overline{A \cap B})) \cap (B \cap (\overline{A \cap B})) = \emptyset$$

$$(A \cap (\overline{A} \cup \overline{B})) \cap (B \cap (\overline{A} \cup \overline{B})) = \emptyset.$$

$$(A \cap \overline{A}) \cup (A \cap \overline{B}) \cap (B \cap \overline{A}) \cup (B \cap \overline{B}) = \emptyset$$

$$(A \cap \overline{B}) \cap (B \cap \overline{A})$$

 $\emptyset - \emptyset$ proved.

$$(b) (A - B) \cup (A \cap B) = A$$

Soln.

$$(A - B) \cup (A \cap B) = A.$$

$$A \cap (\overline{B} \cup B) = A.$$

$$A \cap U = A$$

A = A proved.

$$\therefore A - B = A \cap \overline{B}$$

$$(c) A - (A - B) = A \cap B$$

Pr.

$$A - (A \cap \bar{B}), A \cap B$$

$$A \cap (\overline{A \cap \bar{B}}), A \cap B$$

$$A \cap (\bar{A} \cup B), A \cap B$$

$$(A \cap \bar{A}) \cup (A \cap B), A \cap B$$

$$\emptyset \cup (A \cap B) = A \cap B$$

$$A \cap B \leq A \cap B$$

proved.

$$(d) (\bar{B} \cup (\bar{B} - A)) \leq B$$

Pr.

$$(\bar{B} \cup (\bar{B} \cap \bar{A})) \leq B$$

$$B \cap (\bar{B} \cup (\bar{B} \cap \bar{A})) \leq B$$

$$B \cap \bar{B} \leq B$$

$B \leq B$ proved.